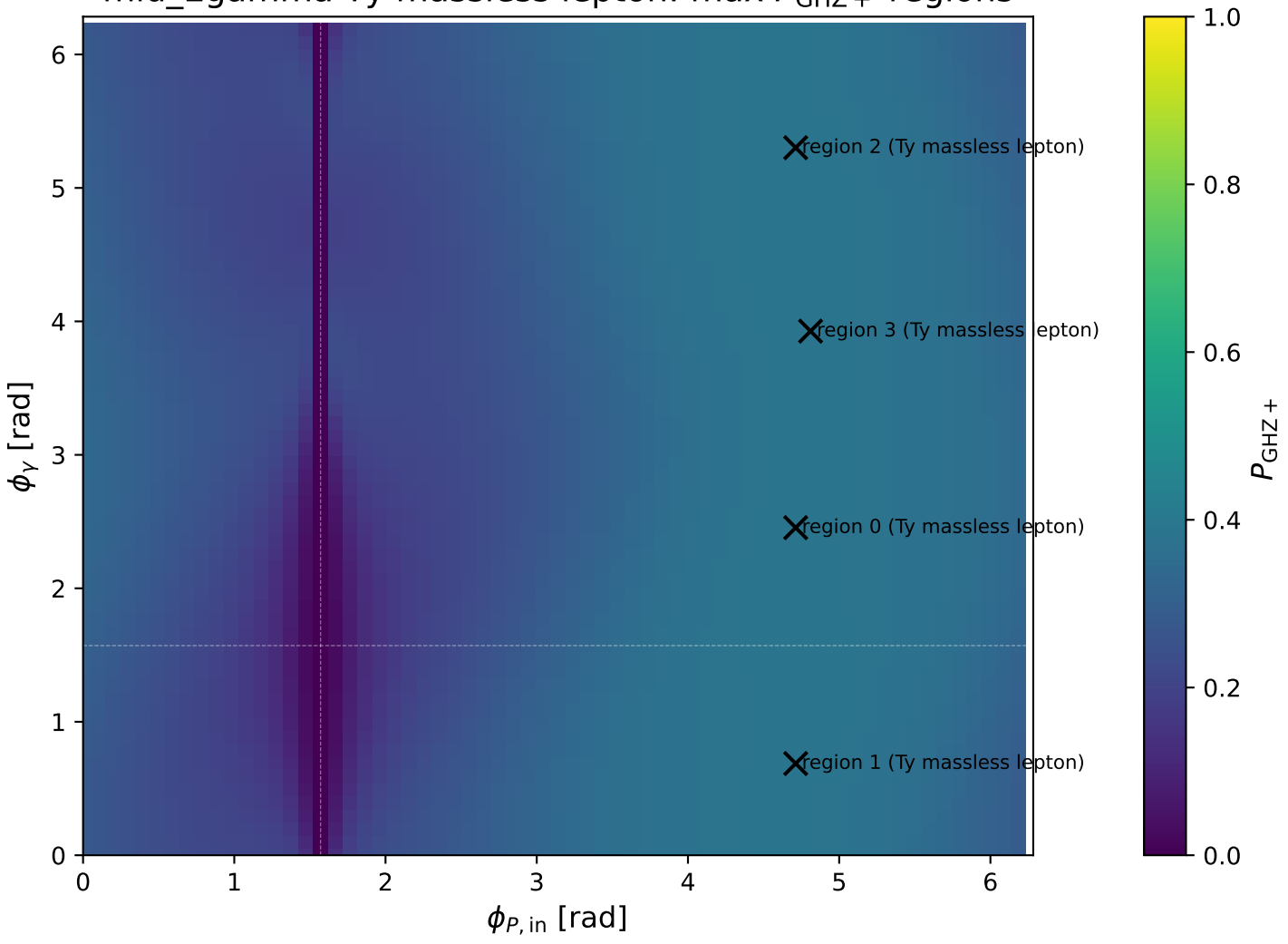
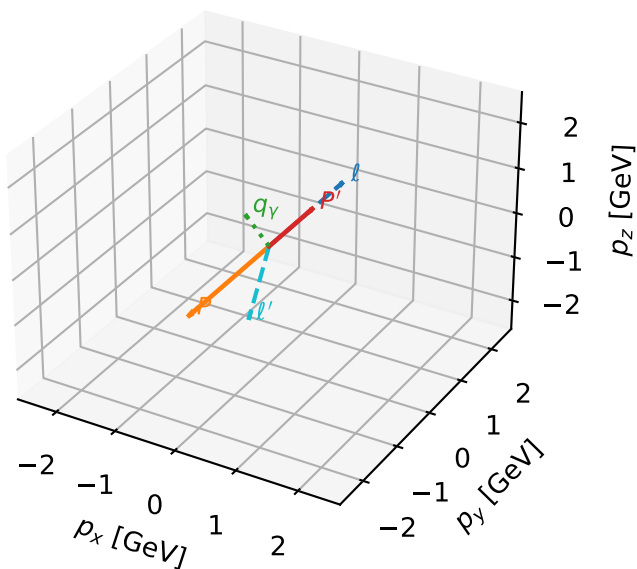


mid\_Egamma Ty massless lepton: max  $P_{\text{GHz}+}$  regions



# Ty massless lepton characteristic max $P_{\text{GHZ}}$ + configuration

3D momenta

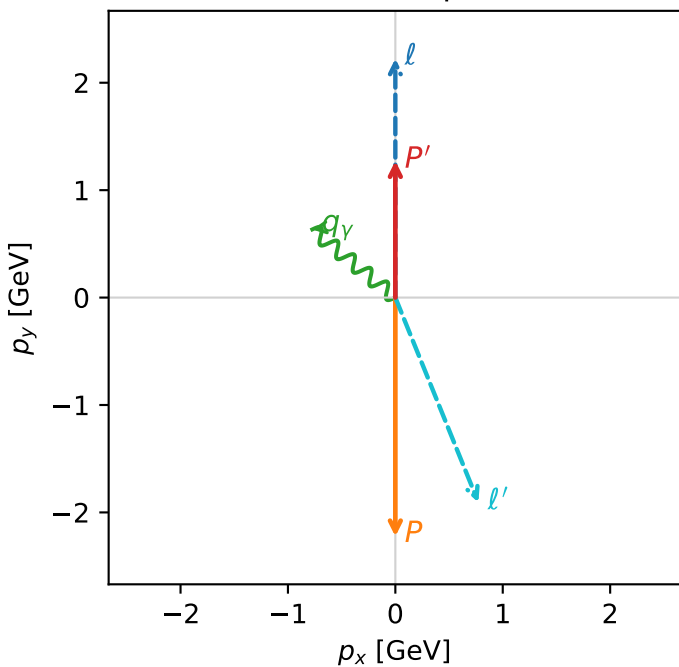


ghz\_purity\_mid\_egamma\_ty\_electron\_region\_0  $P_{\text{GHZ}}=0.384124$   
 region: mid\_Egamma\_Ty\_electron\_region\_0  
 outgoing-state purity: 0.5  
 kinematic point: high\_s\_high\_theta\_in\_mid\_Egamma  
 incoming state:  $\frac{1}{2}\sum_{h_p}\rho(T_{y,\ell_0},h_p)$

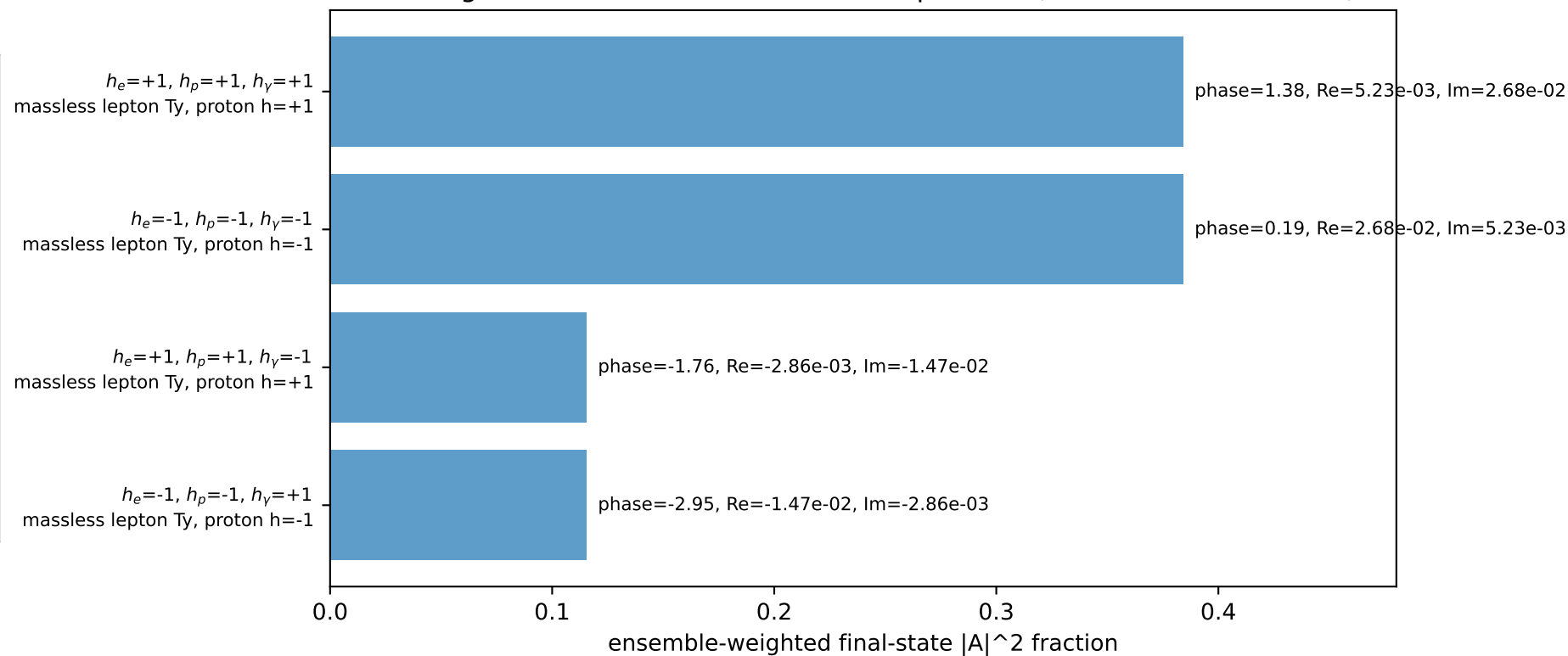
$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=1.27253$   
 $\theta_{\text{in}}=1.57079$ ,  $\phi_\ell=1.5708$ ,  $\phi_P=4.71239$   
 $E_\gamma=1$ ,  $\phi_\gamma=2.45437$   
 $Q^2=17.6377$ ,  $x_B=0.919354$ ,  $t=-11.5344$ ,  $W^2=2.42703$ ,  $y=0.92968$   
 $\theta(\ell',\gamma)=151.441$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E= 2.2244$   $p_x= 0.0000$   $p_y= 2.2244$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= -0.0000$   $p_y= -2.2244$   $p_z= 0.0000$   
 $\ell'$   $E= 2.0576$   $p_x= 0.7730$   $p_y= -1.9069$   $p_z= 0.0000$   
 $P'$   $E= 1.5809$   $p_x= 0.0000$   $p_y= 1.2725$   $p_z= 0.0000$   
 $q_\gamma$   $E= 1.0000$   $p_x= -0.7730$   $p_y= 0.6344$   $p_z= 0.0000$

Transverse plane

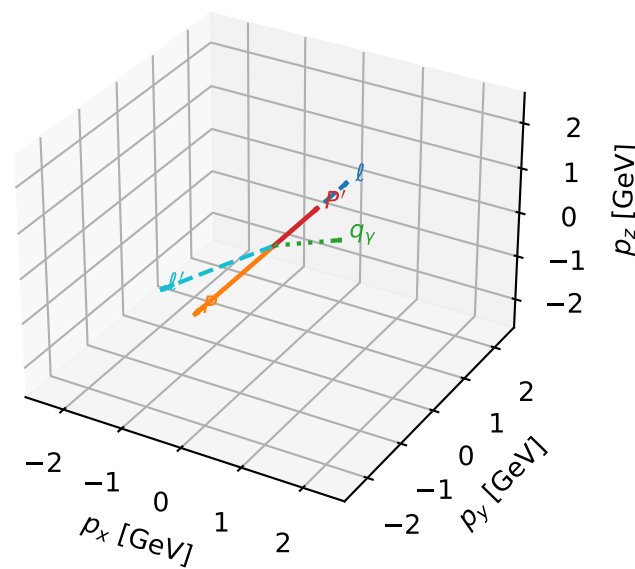


Leading initial-ensemble/final-state components (N=4, retained=99.9%)



# Ty massless lepton characteristic max $P_{\text{GHZ}}$ + configuration

3D momenta

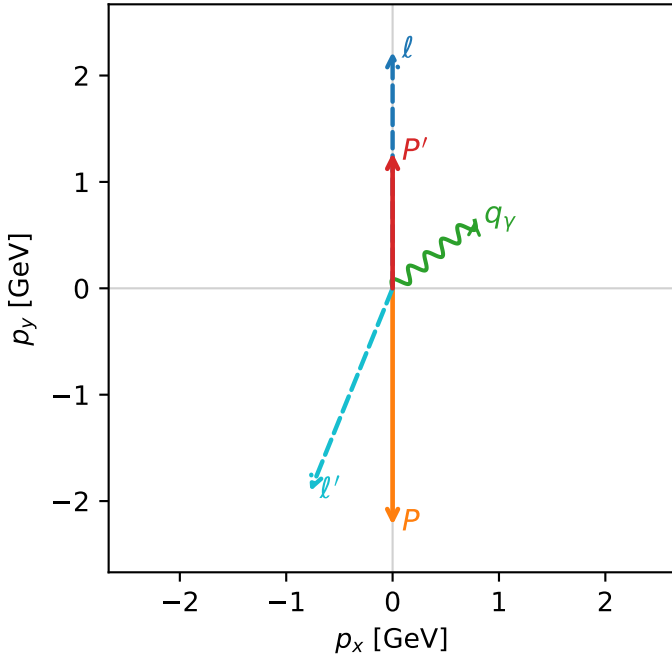


ghz\_purity\_mid\_egamma\_ty\_electron\_region\_1  $P_{\text{GHZ}}=0.384124$   
region: mid\_Egamma\_Ty\_electron\_region\_1  
outgoing-state purity: 0.5  
kinematic point: high\_s\_high\_theta\_in\_mid\_Egamma  
incoming state:  $\frac{1}{2} \sum_{h_p} \rho(T_{y,l_0}, h_p)$

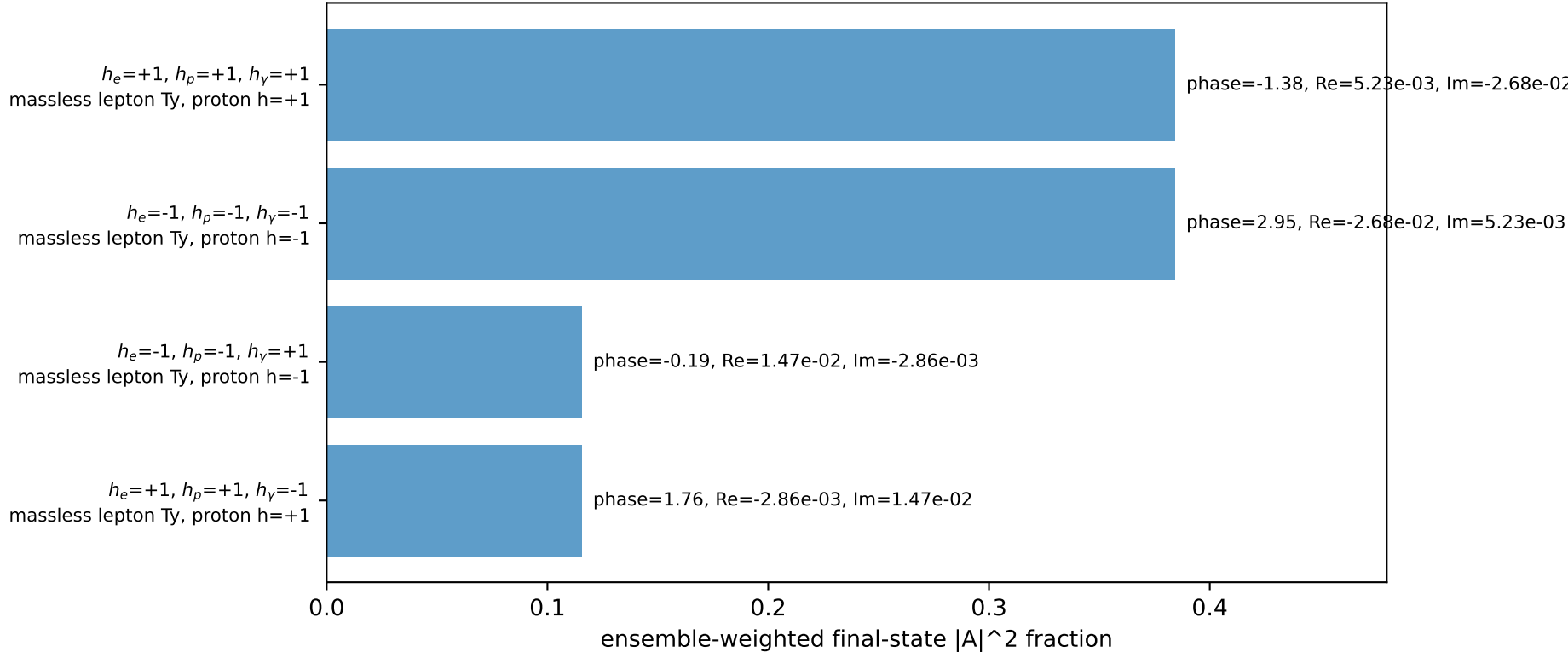
$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=1.27253$   
 $\theta_{\text{in}}=1.57079$ ,  $\phi_l=1.5708$ ,  $\phi_p=4.71239$   
 $E_\gamma=1$ ,  $\phi_\gamma=0.687223$   
 $Q^2=17.6377$ ,  $x_B=0.919354$ ,  $t=-11.5344$ ,  $W^2=2.42703$ ,  $y=0.92968$   
 $\theta(l', \gamma)=151.441$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E= 2.2244$   $p_x= 0.0000$   $p_y= 2.2244$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= -0.0000$   $p_y= -2.2244$   $p_z= 0.0000$   
 $\ell'$   $E= 2.0576$   $p_x= -0.7730$   $p_y= -1.9069$   $p_z= 0.0000$   
 $P'$   $E= 1.5809$   $p_x= 0.0000$   $p_y= 1.2725$   $p_z= 0.0000$   
 $q_\gamma$   $E= 1.0000$   $p_x= 0.7730$   $p_y= 0.6344$   $p_z= 0.0000$

Transverse plane

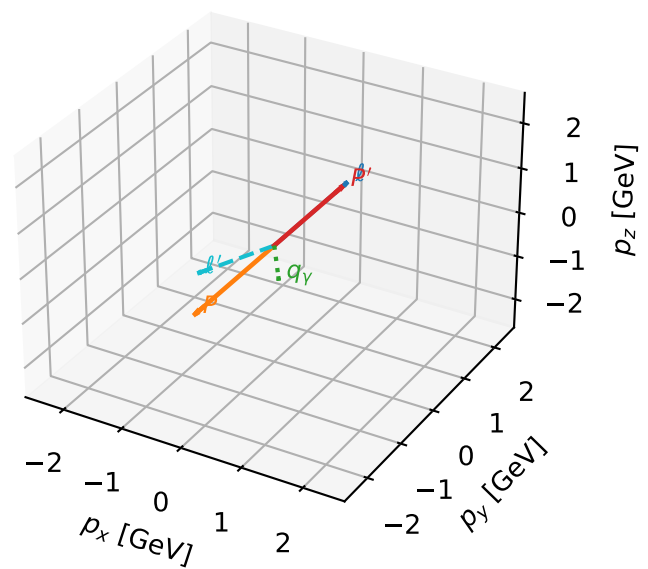


Leading initial-ensemble/final-state components (N=4, retained=99.9%)



# Ty massless lepton characteristic max $P_{\text{GHZ}}$ + configuration

3D momenta

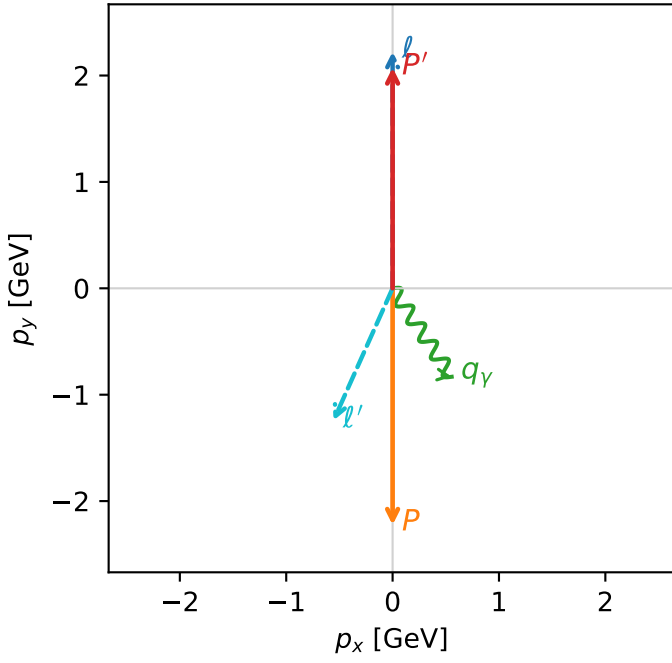


ghz\_purity\_mid\_egamma\_ty\_electron\_region\_2  $P_{\text{GHZ}}=0.383886$   
region: mid\_Egamma\_Ty\_electron\_region\_2  
outgoing-state purity: 0.5  
kinematic point: high\_s\_high\_theta\_in\_mid\_Egamma  
incoming state:  $\frac{1}{2} \sum_{h_p} \rho(T_{y, l_0}, h_p)$

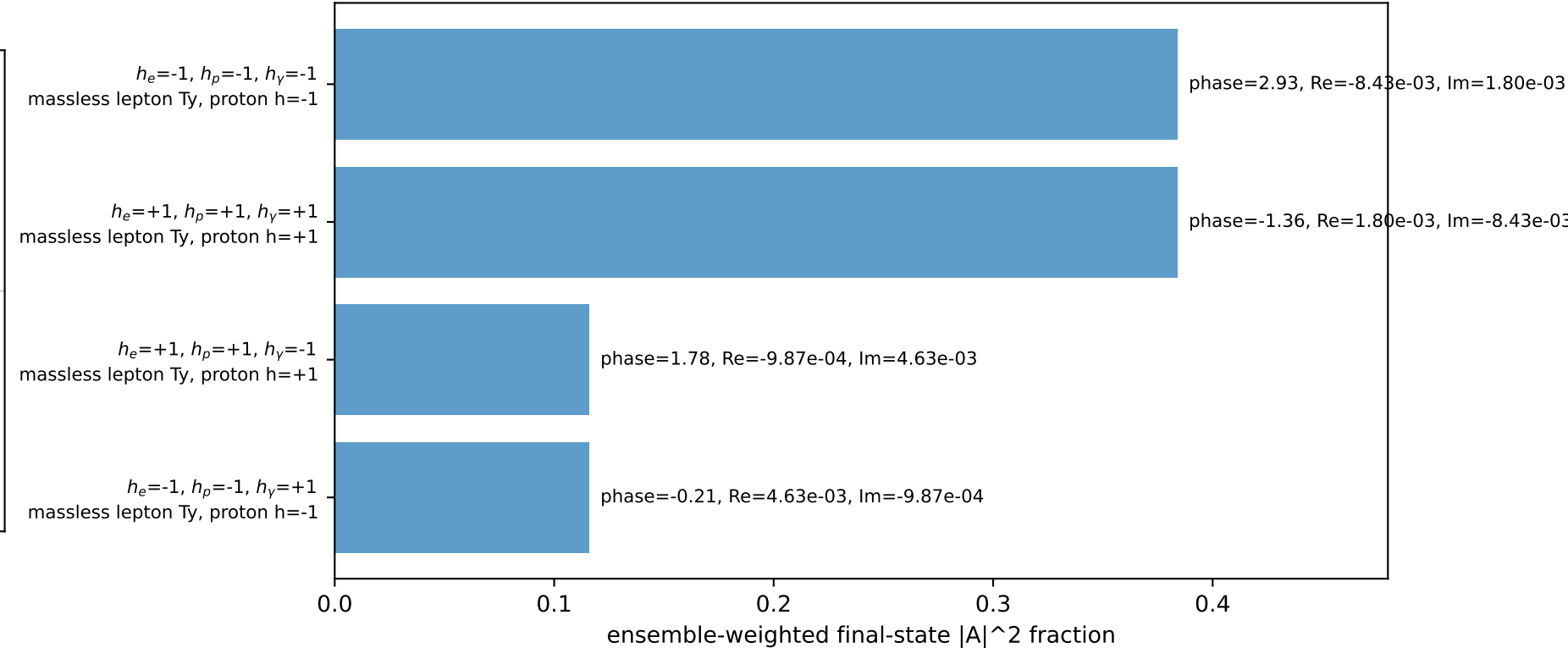
$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=2.07465$   
 $\theta_{\text{in}}=1.57079$ ,  $\phi_l=1.5708$ ,  $\phi_p=4.71239$   
 $E_\gamma=1$ ,  $\phi_\gamma=5.30144$   
 $Q^2=11.5886$ ,  $x_B=0.591487$ ,  $t=-18.4631$ ,  $W^2=8.88355$ ,  $y=0.949422$   
 $\theta(l', \gamma)=57.8296$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E= 2.2244$   $p_x= 0.0000$   $p_y= 2.2244$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= -0.0000$   $p_y= -2.2244$   $p_z= 0.0000$   
 $\ell'$   $E= 1.3617$   $p_x= -0.5556$   $p_y= -1.2432$   $p_z= 0.0000$   
 $P'$   $E= 2.2768$   $p_x= 0.0000$   $p_y= 2.0746$   $p_z= 0.0000$   
 $q_\gamma$   $E= 1.0000$   $p_x= 0.5556$   $p_y= -0.8315$   $p_z= 0.0000$

Transverse plane

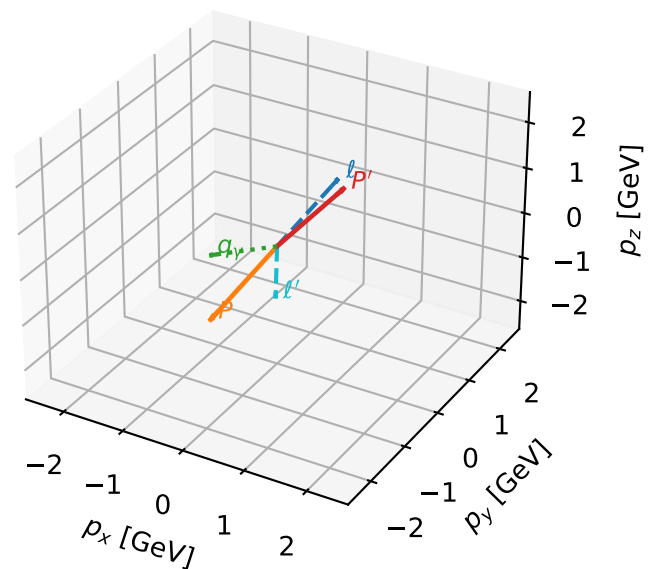


Leading initial-ensemble/final-state components (N=4, retained=99.9%)



# Ty massless lepton characteristic max $P_{\text{GHZ}}$ + configuration

3D momenta

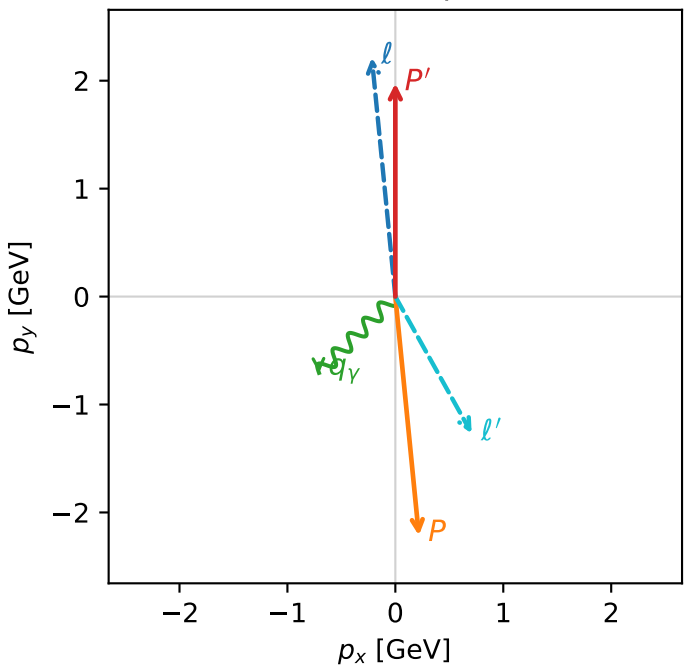


ghz\_purity\_mid\_egamma\_ty\_electron\_region\_3  $P_{\text{GHZ}}=0.383831$   
 region: mid\_Egamma\_Ty\_electron\_region\_3  
 outgoing-state purity: 0.5  
 kinematic point: high\_s\_high\_theta\_in\_mid\_Egamma  
 incoming state:  $\frac{1}{2} \sum_{h_p} \rho(T_{y,\ell_0}, h_p)$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=1.9752$   
 $\theta_{\text{in}}=1.57079$ ,  $\phi_\ell=1.66897$ ,  $\phi_P=4.81056$   
 $E_\gamma=1$ ,  $\phi_\gamma=3.92699$   
 $Q^2=12.382$ ,  $x_B=0.633397$ ,  $t=-17.5427$ ,  $W^2=8.0464$ ,  $y=0.947304$   
 $\theta(\ell', \gamma)=74.1447$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E= 2.2244$   $p_x= -0.2180$   $p_y= 2.2137$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= 0.2180$   $p_y= -2.2137$   $p_z= 0.0000$   
 $\ell'$   $E= 1.4519$   $p_x= 0.7071$   $p_y= -1.2681$   $p_z= 0.0000$   
 $P'$   $E= 2.1866$   $p_x= 0.0000$   $p_y= 1.9752$   $p_z= 0.0000$   
 $q_\gamma$   $E= 1.0000$   $p_x= -0.7071$   $p_y= -0.7071$   $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=4, retained=99.9%)

